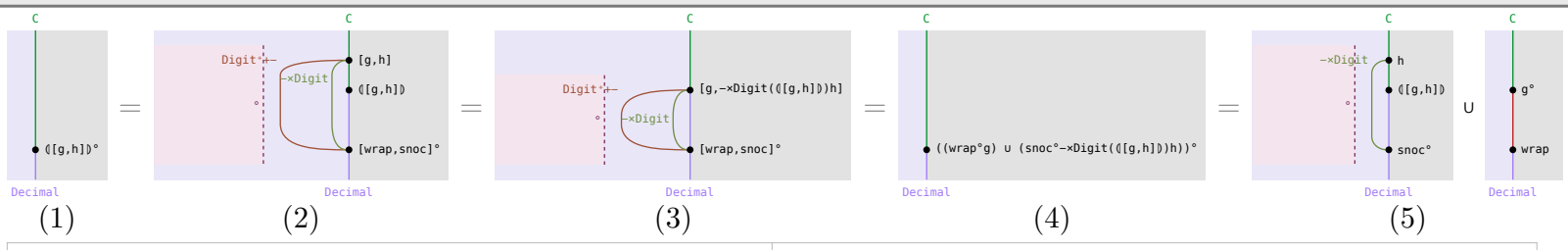
Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

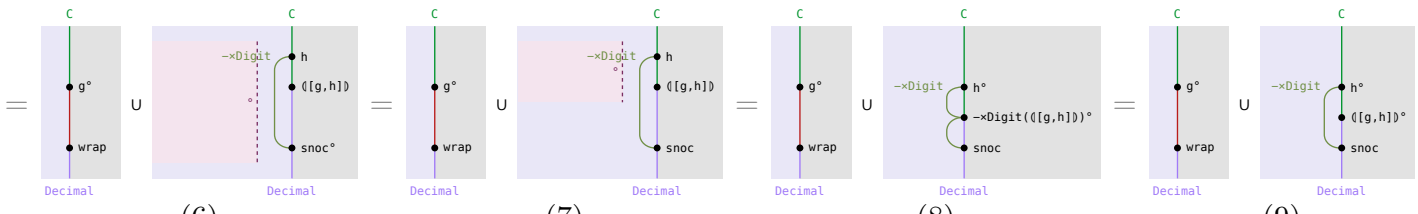
`⦇[g,h]⦈° = (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)` (6.1a)

the converse of a fold over decimals undoes the algebra `[g,h]` and then either `wrap`, or the fold's converse on the front and `snoc`



(1) (2) (3) (4) (5)

1	`⦇[g,h]⦈°`	`⦇φ⦈ = [wrap,snoc]° Digit⁺+−(−×Digit(⦇φ⦈)) φ`
2	`= ([wrap,snoc]° Digit⁺+−(−×Digit(⦇[g,h]⦈))) [g,h])°`	`(R + S) [P,Q] = [R P,S Q]`
3	`= ([wrap,snoc]° [g,−×Digit(⦇[g,h]⦈) h])°`	`[U,V]° [R,S] = (U° R) ∪ (V° S)`
4	`= ((wrap° g) ∪ (snoc° −×Digit(⦇[g,h]⦈) h))°`	`(R ∪ S)° = S° ∪ R°`
5	`= (snoc° −×Digit(⦇[g,h]⦈) h)° ∪ (wrap° g)°`	`R ∪ S = S ∪ R`



(6) (7) (8) (9)

6	`= (wrap° g)° ∪ (snoc° −×Digit(⦇[g,h]⦈) h)°`	`(R S)° = S° R°`
7	`= (g° wrap) ∪ ((−×Digit(⦇[g,h]⦈) h)° snoc)`	`(R h)° = h° R°`
8	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈)° snoc)`	`−×Digit(R)° = −×Digit(R°)`
9	`= (g° wrap) ∪ (h° −×Digit(⦇[g,h]⦈°) snoc)`	

`val° = (embed° wrap) ∪ (op° −×Digit(val°) snoc)`

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(6.1b)

1	`op°(m,p) ⇔ π₁(p) = m/10 ∧ π₂(p) = m % 10` `op°` splits a number into its quotient and remainder by 10
2	`(∃p. 0 < π₁(p) ∧ op°(m,p)) ⇔ 10 ≤ m` `op°` gives a pair with a nonzero first component exactly at the numbers with two or more digits
3	`(∃d. embed°(m,d)) ⇔ 0 < m ∧ m < 10` `embed°` gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

`Monotonic(φ) ≜ ∀X. ∀Y. X ⊑ Y ⇒ φ(X) ⊑ φ(Y)` — φ maps the hom-set `A↛B` into itself with `X⊑Y⇒φ(X)⊑φ(Y)`.

`(μX : φ(X))` is the least `X : A↛B` with `φ(X)⊑X`.

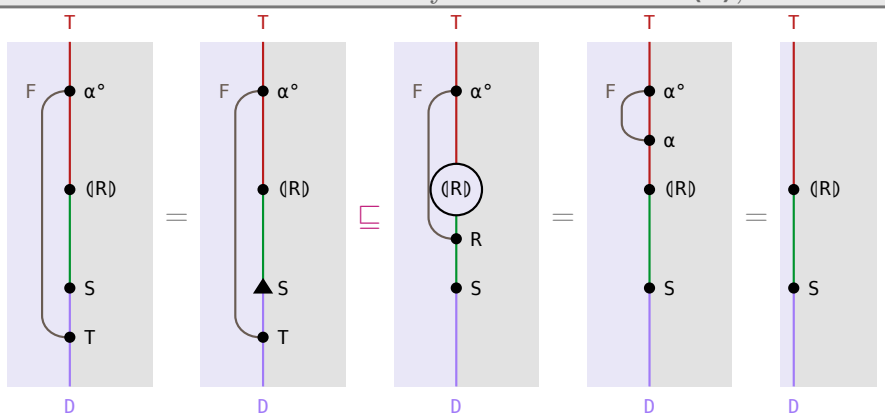
(6.2b)

	the law	what it says
1	`φ(Y) ⊑ Y ⇒ (μX : φ(X)) ⊑ Y`	to bound `(μX : φ(X))` above, exhibit one `Y` the body does not grow past — the half §Section 6.3. and every chapter after it uses
2	`Monotonic(φ) ⇒ φ(μX : φ(X)) = (μX : φ(X))`	**Knaster–Tarski**: the least solution of `φ(X)⊑X` already solves `φ(X)=X`, so the least prefix point and the least fixed point are one relation
3	`α° F(X) R ⊑ X ⇒ (R) ⊑ X` (6.2)	a relation `X` that contains one unfolding of the fold's recursion at `X` contains the fold
4	`X ⊑ α° F(X) R ⇒ X ⊑ (R)` (6.3)	a relation `X` contained in one unfolding of the fold's recursion at `X` is contained in the fold

(6.2c)

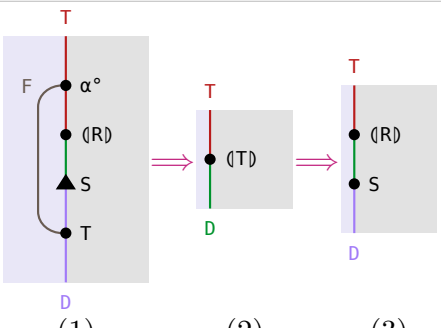
`F(S) T ⊑ R S`
`⇒ (T) ⊑ (R) S`

if **S** followed by **R** absorbs **T** after **F(S)**, the fold of **T** is below the fold of **R** followed by **S**



(1) (2) (3) (4) (5)

1	`α° F((R) S) T`	
2	`= α° F((R)) F(S) T`	`F(S) T ⊑ R S`
3	`⊑ α° F((R)) R S`	`α (R) = F((R)) R`
4	`= α° α (R) S`	`α° α = 𝟙`
5	`= (R) S`	

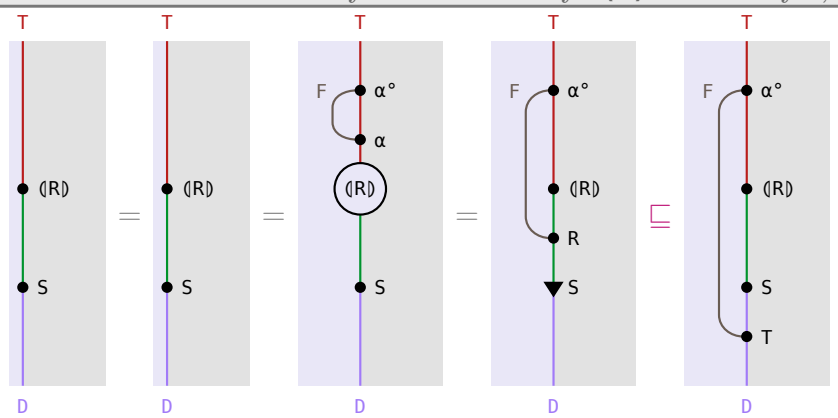


(1) (2) (3)

1	`F(S) T ⊑ R S`	`F(S) T ⊑ R S` `⇒ α° F((R) S) T ⊑ (R) S`
2	`⟹ α° F((R) S) T ⊑ (R) S`	`α° F(X) T ⊑ X` `⇒ (T) ⊑ X`
3	`⟹ (T) ⊑ (R) S`	

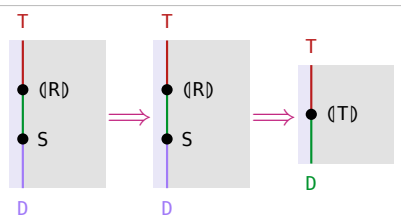
$$R\ S \sqsubseteq F(S)\ T \quad \Rightarrow (\!|R|\!)\ S \sqsubseteq (\!|T|\!) \tag{6.2d}$$

if R followed by S is absorbed by $F(S)$ followed by T, the fold of R followed by S is below the fold of T



(1) (2) (3) (4) (5)

1	$(\!|R|\!)\ S$	
2	$= (\!|R|\!)\ S$	$\alpha^\circ\ \alpha = \mathbb{1}$
3	$= \alpha^\circ\ \alpha\ (\!|R|\!)\ S$	$\alpha\ (\!|R|\!) = F(\!|R|\!)\ R$
4	$= \alpha^\circ\ F(\!|R|\!)\ R\ S$	$R\ S \sqsubseteq F(S)\ T$
5	$\sqsubseteq \alpha^\circ\ F(\!|R|\!)\ S)\ T$	



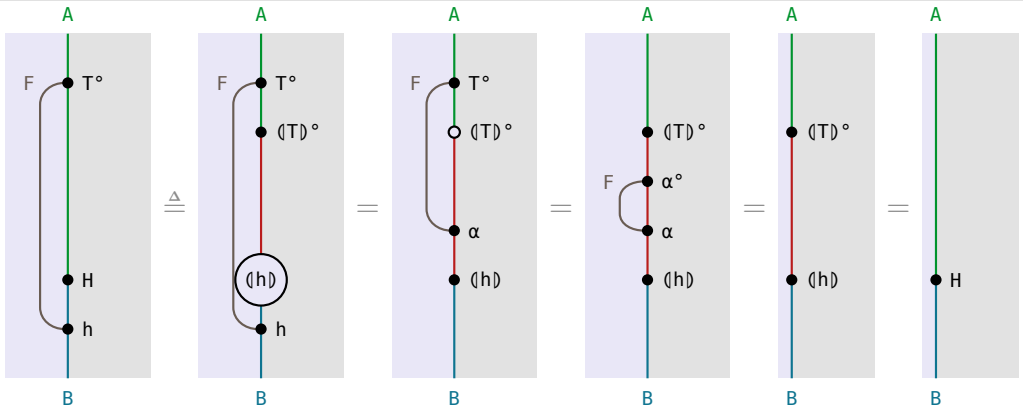
(1) (2) (3)

1	$R\ S \sqsubseteq F(S)\ T$	$R\ S \sqsubseteq F(S)\ T$ $\Rightarrow (\!|R|\!)\ S \sqsubseteq \alpha^\circ\ F(\!|R|\!)\ S)\ T$
2	$\Longrightarrow (\!|R|\!)\ S \sqsubseteq \alpha^\circ\ F(\!|R|\!)\ S)\ T$	$X \sqsubseteq \alpha^\circ\ F(X)\ T$ $\Rightarrow X \sqsubseteq (\!|T|\!)$
3	$\Longrightarrow (\!|R|\!)\ S \sqsubseteq (\!|T|\!)$	

§6.3. Hylomorphisms

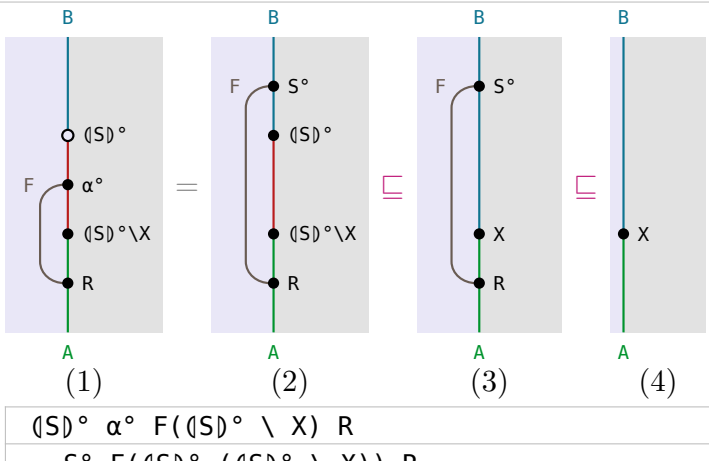
⦇S⦈° ⦇R⦈ = (μX : S° F(X) R) (6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation



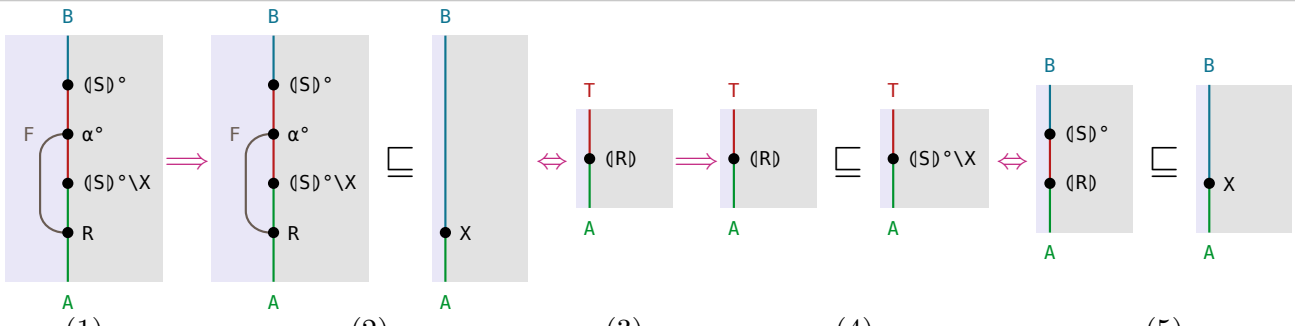
(1) (2) (3) (4) (5) (6)

1	T° F(H) h	H ≜ ⦇T⦈° ⦇h⦈
2	≜ T° F(⦇T⦈° ⦇h⦈) h	α ⦇h⦈ = F(⦇h⦈) h
3	= T° F(⦇T⦈°) α ⦇h⦈	⦇T⦈° α° = T° F(⦇T⦈°)
4	= ⦇T⦈° α° α ⦇h⦈	α° α = 𝟙
5	= ⦇T⦈° ⦇h⦈	H = ⦇T⦈° ⦇h⦈
6	= H	



(1) (2) (3) (4)

1	⦇S⦈° α° F(⦇S⦈° \ X) R	⦇S⦈° α° = S° F(⦇S⦈°)
2	= S° F(⦇S⦈° (⦇S⦈° \ X)) R	S (S \ X) ⊑ X
3	⊑ S° F(X) R	S° F(X) R ⊑ X
4	⊑ X	

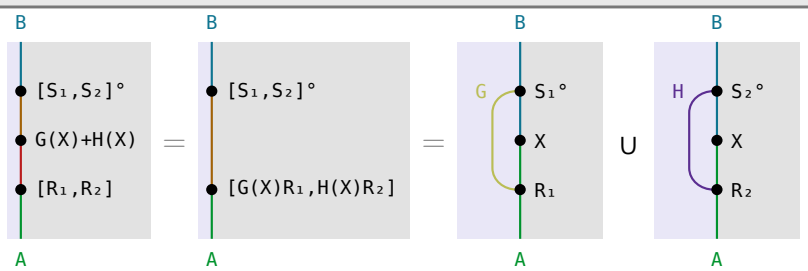


(1) (2) (3) (4) (5)

1	S° F(X) R ⊑ X	S° F(X) R ⊑ X ⇒ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X
2	⟹ ⦇S⦈° α° F(⦇S⦈° \ X) R ⊑ X	X ⊑ S \ R ⇔ S X ⊑ R
3	⟺ α° F(⦇S⦈° \ X) R ⊑ ⦇S⦈° \ X	α° F(X) R ⊑ X ⇒ ⦇R⦈ ⊑ X
4	⟹ ⦇R⦈ ⊑ ⦇S⦈° \ X	X ⊑ S \ R ⇔ S X ⊑ R
5	⟺ ⦇S⦈° ⦇R⦈ ⊑ X	

⦇[S₁,S₂]⦈° ⦇[R₁,R₂]⦈ = (μX : (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)) (6.3b)

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results



(1) (2) (3)

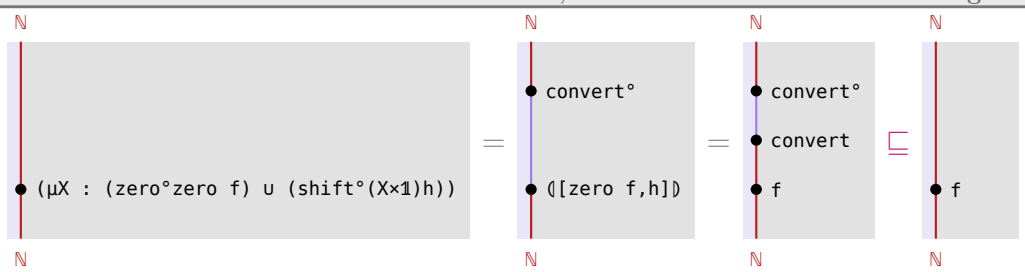
1	`[S₁,S₂]° (G(X) + H(X)) [R₁,R₂]`	`(R + S) [R₁,R₂] = [R R₁,S R₂]`
2	`= [S₁,S₂]° [G(X) R₁,H(X) R₂]`	`[S₁,S₂]° [R,S] = (S₁° R) ∪ (S₂° S)`
3	`= (S₁° G(X) R₁) ∪ (S₂° H(X) R₂)`	

§6.4. Fast exponentiation and modulus computation

shift f = (f × 𝟙) h (6.4a)

⇒ (μX : (zero° zero f) ∪ (shift° (X × 𝟙) h)) ⊑ f

once **f** turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



1	(μX : (zero° zero f) ∪ (shift° (X × 𝟙) h))	convert° ⦇[g,h]⦈ = (μX : (zero° g) ∪ (shift° (X × 𝟙) h))
2	= convert° ⦇[zero f,h]⦈	zero f = g; shift f = (f × 𝟙) h ⇒ convert f = ⦇[g,h]⦈
3	= convert° convert f	Simple(convert)
4	⊑ f	

(μX : (zero° one) ∪ (shift° (X × 𝟙) op(a))) ⊑ exp(a) (6.4b)

the recursion that halves the exponent at each step computes **a** to the power **b**

`zero exp(a) = one`

`a` to the power `0` is `1`

`shift exp(a) = (exp(a) × 𝟙) op(a)`

`a` to the power `2n+d` is `op(a)` of `a` to the power `n` and `d`

(μX : (zero° zero) ∪ (shift° (X × 𝟙) op(b))) ⊑ mod(b) (6.4c)

the recursion that halves **a** at each step computes the remainder of **a** by **b**

`zero mod(b) = zero`

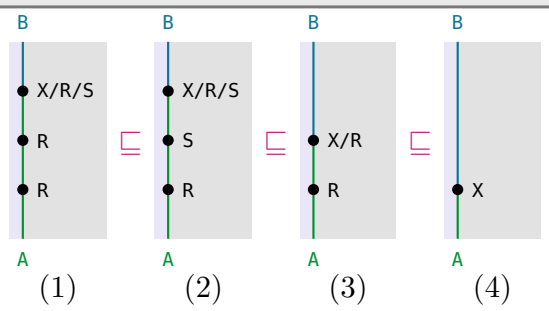
`0 mod b` is `0`

`shift mod(b) = (mod(b) × 𝟙) op(b)`

`(2a+d) mod b` is `op(b)` of `a mod b` and `d`

§6.5. Unique fixed points

$$R^2 \sqsubseteq S\ R \quad (6.5a)$$
$$\Rightarrow (X\ /\ R\ /\ S)\ R^2 \sqsubseteq X$$
((X/R)/S) followed by RR stays below X when RR⊑SR
inductive(S); $R^2 \sqsubseteq S\ R$
⇒ inductive(R)
R is inductive when RR⊑SR for an inductive S



1	(X / R / S) R^2	$R^2 \sqsubseteq S\ R$
2	$\sqsubseteq$ (X / R / S) S R	(R / S) S ⊑ R
3	$\sqsubseteq$ (X / R) R	(X / R) R ⊑ X
4	$\sqsubseteq$ X	

§6.5.1. Membership

member = ∅ (6.5.1a)
a constant relator records no elements, so its membership is empty
member = [member,member]
a member of F+G is a member of whichever summand the value is in
member = (π_1 member) ∪ (π_2 member)
a member of F×G is a member of either component

§6.5.2. Consequences

inductive(S member); X = S F(X) R; Y = S F(Y) R (6.5.2a)
⇒ X = Y
when S member is inductive, two solutions X, Y of X=SF(X)R are equal

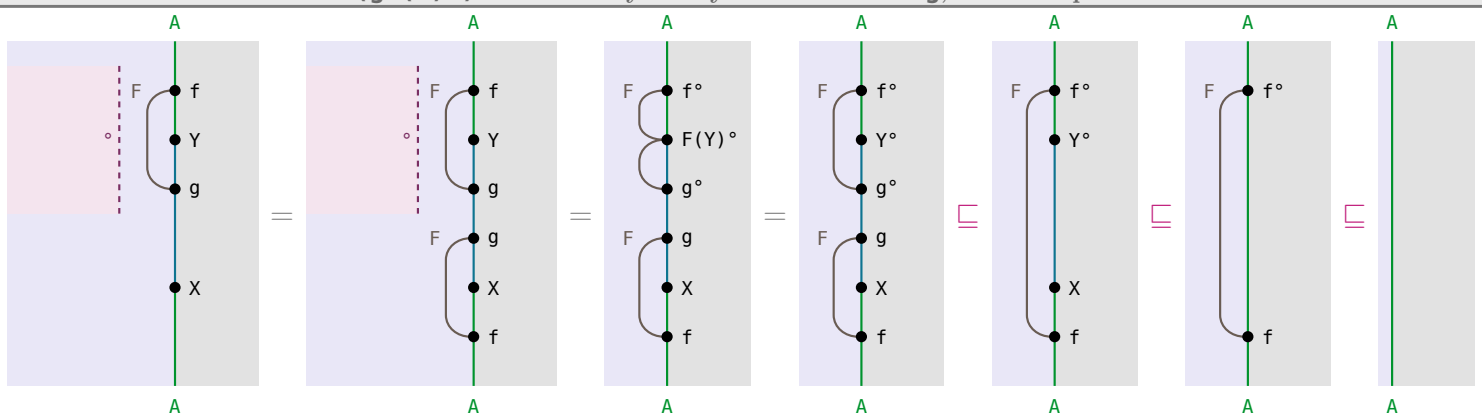
inductive(S member); Entire(S); Entire(R); S F(X) R ⊑ X (6.5.2b)
⇒ Entire(X)
when S member is inductive and S, R are entire, a pre-fixed point of X↦SF(X)R is entire

inductive(g member); Map(g); Map(f); X = g F(X) f
⇒ Map(X) (6.5.2c)

when **g member** is inductive and **f**, **g** are maps, the solution of **X=gF(X)f** is a map

X = g F(X) f; Simple(g); Simple(f); Y° X ⊑ 𝟙
⇒ (g F(Y) f)° X ⊑ 𝟙

(gF(Y)f)° followed by **X** stays below **𝟙** when **g**, **f** are simple and **Y°X⊑𝟙**

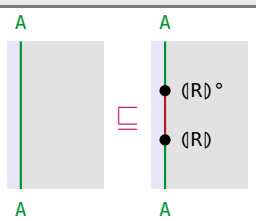


(1) (2) (3) (4) (5) (6) (7)

1	(g F(Y) f)° X	X = g F(X) f
2	= (g F(Y) f)° g F(X) f	(R S)° = S° R°
3	= f° F(Y)° g° g F(X) f	PreservesRecip(F)
4	= f° F(Y°) g° g F(X) f	Simple(g)
5	⊑ f° F(Y° X) f	Y° X ⊑ 𝟙
6	⊑ f° f	Simple(f)
7	⊑ 𝟙	

inductive(R° member); 𝟙 ⊑ R° R
⇒ 𝟙 ⊑ (R)° (R) (6.5.2d)

when **R°member** is inductive and **R** is surjective, the fold **(R)** is surjective

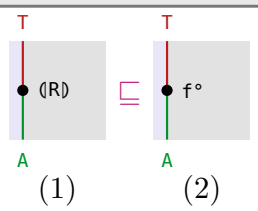


(1) (2)

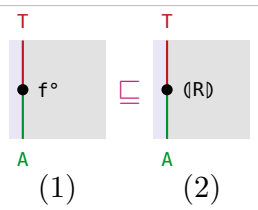
1	𝟙	inductive(S member); Entire(S); Entire(R); S F(X) R ⊑ X ⇒ Entire(X) at S=R°
2	⊑ (R)° (R)	

Map(f); R f ⊑ F(f) α; 𝟙 ⊑ R° R
⇒ f° = ⦇R⦈ (6.5.2e)

a map f with Rf⊑F(f)α, for a surjective R, has the fold ⦇R⦈ as its converse



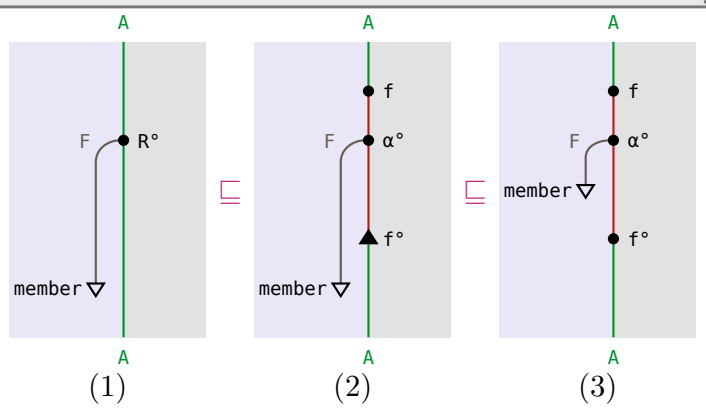
1	⦇R⦈	shunting f, fusion, Rf⊑F(f)α
2	⊑ f°	



1	f°	inductive(R° member); 𝟙 ⊑ R° R ⇒ 𝟙 ⊑ ⦇R⦈° ⦇R⦈ and ⦇R⦈f⊑𝟙
2	⊑ ⦇R⦈	

Map(f); R f ⊑ F(f) α
⇒ R° member ⊑ f α° member f° (6.5.2f)

R°member is below α°member conjugated by f, so it is inductive when α°member is



1	R° member	Map(f); R f ⊑ F(f) α ⇒ R° ⊑ f α° F(f°)
2	⊑ f α° F(f°) member	F(R) member ⊑ member R
3	⊑ f α° member f°	

§6.6. Sorting by selection

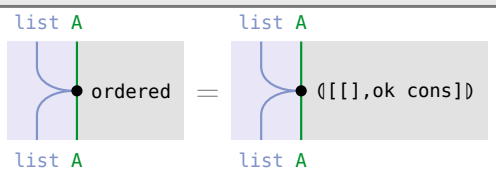
§6.6.1. The function sort

(6.6.1a)

	definition	type	meaning
1	`ordered[] = true` `ordered([a]#xs) = (∀x. x ∈ xs ⇒ R(a,x)) ∧ ordered(xs)`	`[A]→ [A]`	every element is R-below each element after it
2	`(a,xs) ok q ⇔ (a,xs) = q ∧ (∀x. x ∈ xs ⇒ R(a,x))`	`A×[A]→ A×[A]`	the coreflexive passing `(a, x)` when `a` is R-below every element of `x`

`ordered = ⦇[[],ok cons]⦈` (6.6.1b)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

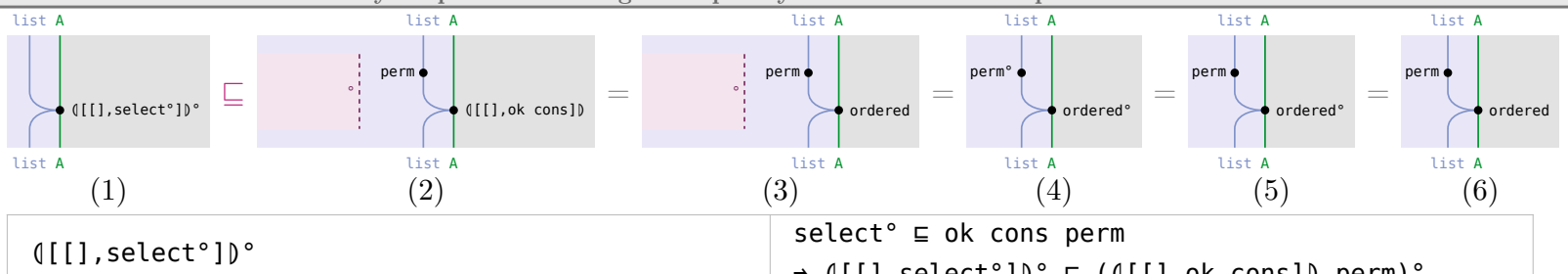


1	`ordered`	fold uniqueness
2	`= ⦇[[],ok cons]⦈`	

§6.6.2. Selection sort

`select° ⊑ ok cons perm`
`⇒ ⦇[[],select°]⦈° ⊑ perm ordered` (6.6.2a)

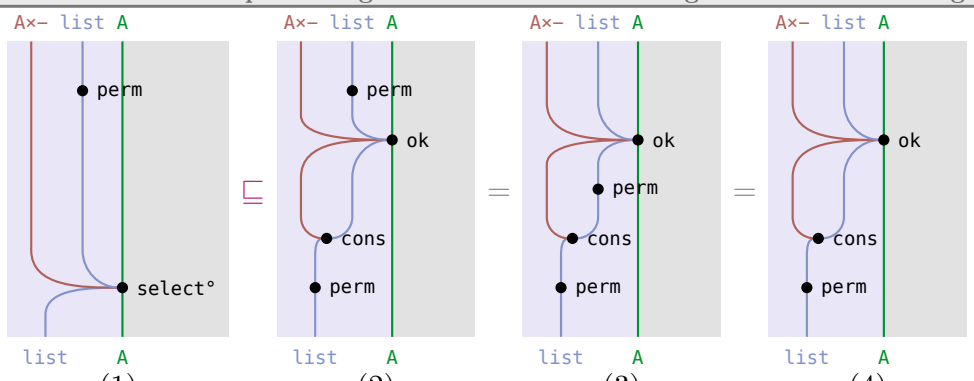
every output of unfolding the input by **select** is a sorted permutation of it



1	`⦇[[],select°]⦈°`	`select° ⊑ ok cons perm` `⇒ ⦇[[],select°]⦈° ⊑ (⦇[[],ok cons]⦈ perm)°`
2	`⊑ (⦇[[],ok cons]⦈ perm)°`	(6.6.1b) `ordered = ⦇[[],ok cons]⦈`
3	`= (ordered perm)°`	`(ordered perm)° = perm° ordered°`
4	`= perm° ordered°`	`perm° = perm`
5	`= perm ordered°`	`ordered° = ordered`
6	`= perm ordered`	

`select° ⊑ ok cons perm`
`⇒ (𝟙 × perm) select° ⊑ ok cons perm` (6.6.2b)

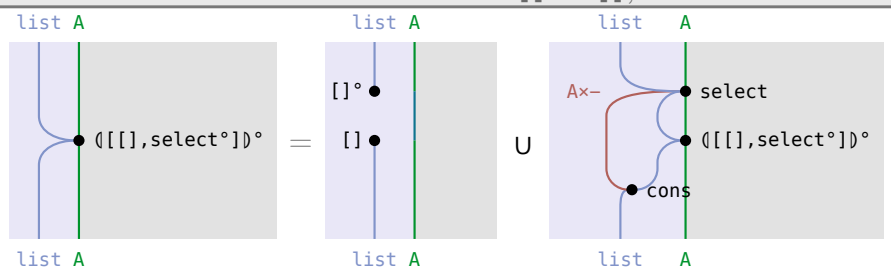
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



1	`(𝟙 × perm) select°`	`select° ⊑ ok cons perm`
2	`⊑ (𝟙 × perm) ok cons perm`	`ok (𝟙 × perm) = (𝟙 × perm) ok`
3	`= ok (𝟙 × perm) cons perm`	`cons perm = (𝟙 × perm) cons perm`
4	`= ok cons perm`	

(𝕝[[],select°]𝕝)° = ([]° []) ∪ (select (𝟙 × 𝕝[[],select°]𝕝°) cons) (6.6.2c)

selection sort returns `[]` on `[]`, and otherwise selects `(a,y)`, sorts `y` and conses `a` back on



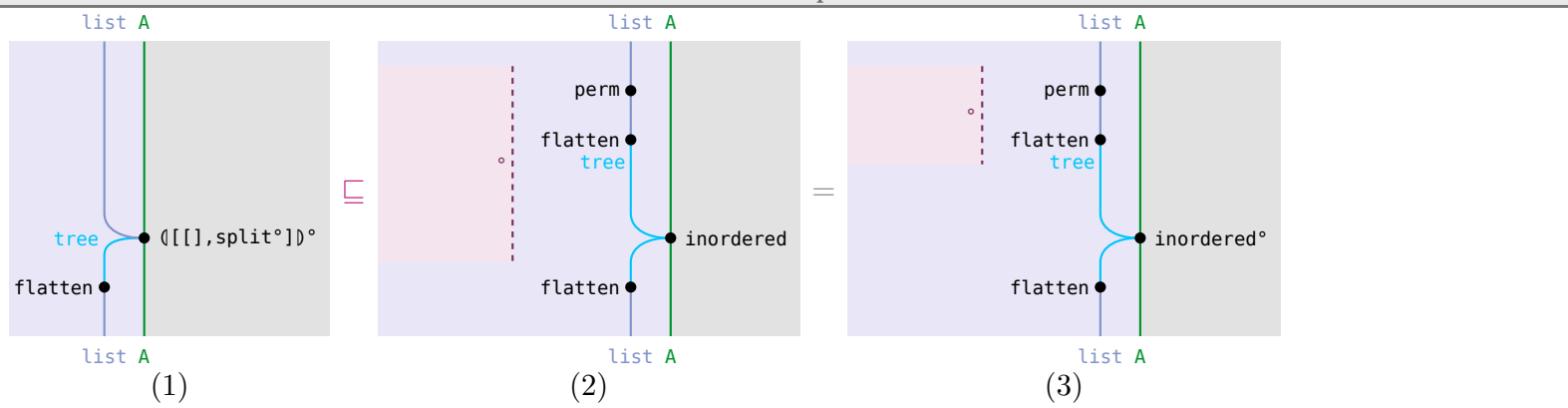
(1) (2)

1	`𝕝[[],select°]𝕝°`	unfolding the converse of a fold
2	`= ([]° []) ∪ (select (𝟙 × 𝕝[[],select°]𝕝°) cons)`	

§6.6.3. Quicksort

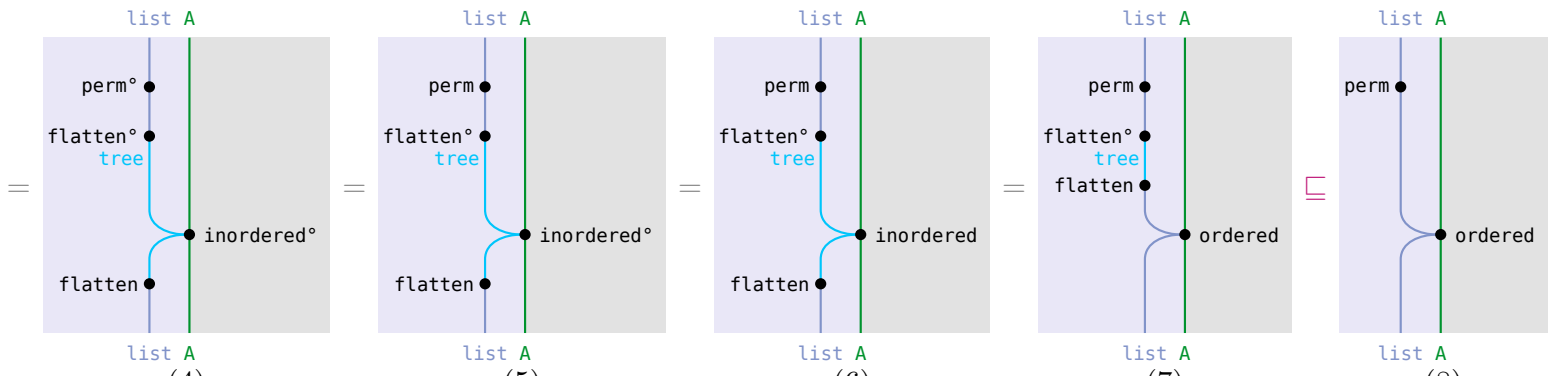
(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)); split° ⊑ check' join perm ⇒ 𝕝[[],split°]𝕝° flatten ⊑ perm ordered (6.6.3a)

every output of unfolding the input by `split` into a tree and flattening that tree is a sorted permutation of the input



(1) (2) (3)

1	`𝕝[[],split°]𝕝° flatten`	`split° ⊑ check' join perm ⇒ 𝕝[[],split°]𝕝° flatten ⊑ (inordered flatten perm)° flatten`
2	`⊑ (inordered flatten perm)° flatten`	`(inordered S)° = S° inordered°`
3	`= (flatten perm)° inordered° flatten`	`(flatten perm)° = perm° flatten°`

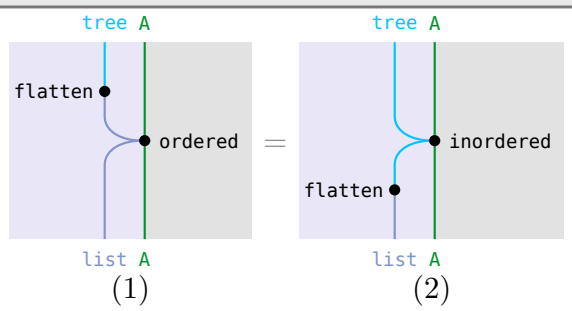


(4) (5) (6) (7) (8)

4	`= perm° flatten° inordered° flatten`	`perm° = perm`
5	`= perm flatten° inordered° flatten`	`inordered° = inordered`
6	`= perm flatten° inordered flatten`	(6.6.3b) `(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) ⇒ flatten ordered = inordered flatten`
7	`= perm flatten° flatten ordered`	`flatten° flatten ⊑ 𝟙`
8	`⊑ perm ordered`	

(∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c))
⇒ flatten ordered = inordered flatten (6.6.3b)

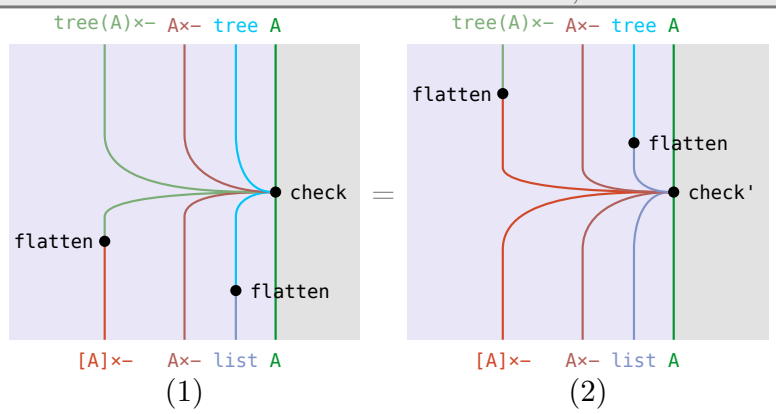
flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening



1	`flatten ordered`	R transitive: a flattened tree is ordered iff every node passes `check`
2	`= inordered flatten`	

check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check' (6.6.3c)

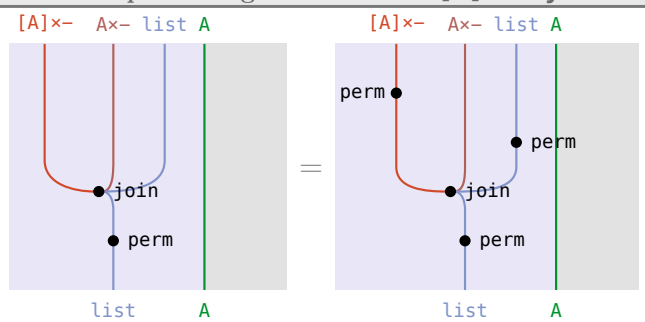
testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening



1	`check (flatten × (𝟙 × flatten))`	`b intree x ⇔ b inlist flatten(x)`
2	`= (flatten × (𝟙 × flatten)) check'`	

join perm = (perm × (𝟙 × perm)) join perm (6.6.3d)

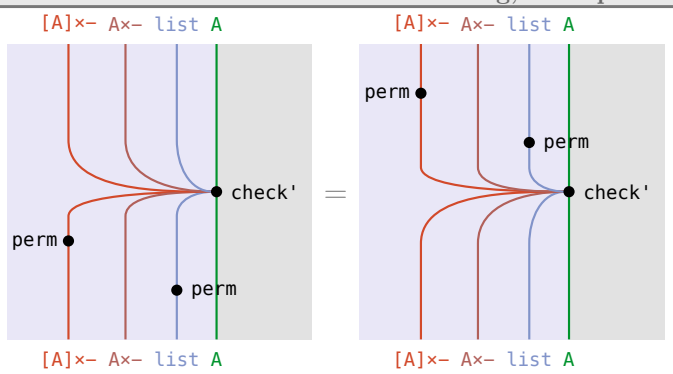
permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list



1	`join perm`	`perm` is transitive and closed under `++`
2	`= (perm × (𝟙 × perm)) join perm`	

check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check' (6.6.3e)

testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged

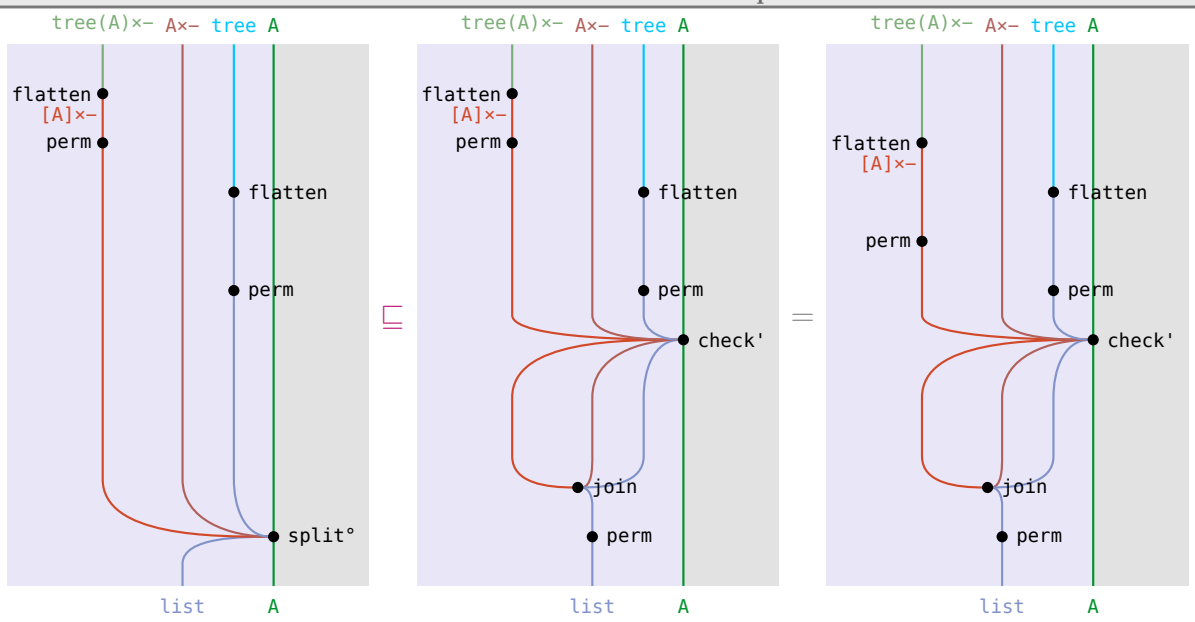



```
1  check' (perm × (𝟙 × perm))          b inlist x ⇔ b inlist perm(x)
2  = (perm × (𝟙 × perm)) check'
```

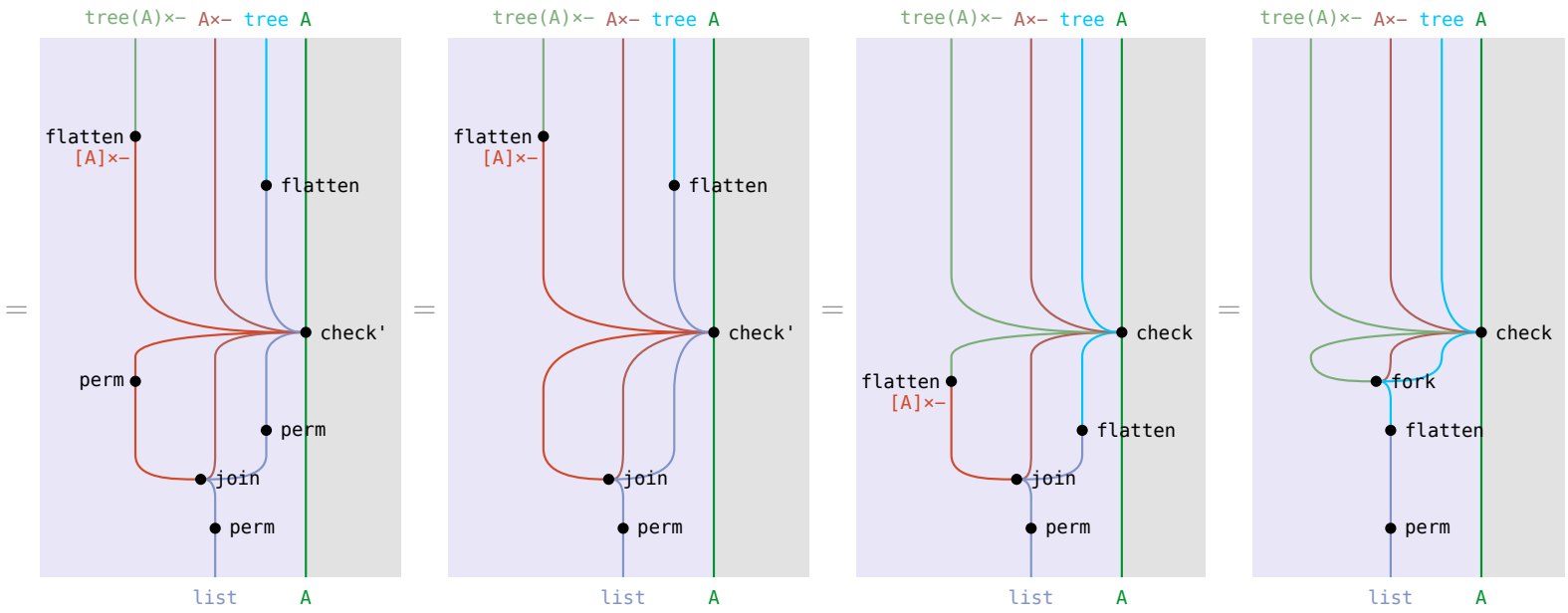
split° ⊑ check' join perm **(6.6.3f)**

⇒ ((flatten perm) × (𝟙 × (flatten perm))) split° ⊑ check fork flatten perm

undoing **split** and then flattening and permuting both parts lands among the **check**ed forks whose flattening is permuted



1	((flatten perm) × (𝟙 × (flatten perm))) split°	split° ⊑ check' join perm
2	⊑ ((flatten perm) × (𝟙 × (flatten perm))) check' join perm	
3	= (flatten × (𝟙 × flatten)) (perm × (𝟙 × perm)) check' join perm	(6.6.3e) check' (perm × (𝟙 × perm)) = (perm × (𝟙 × perm)) check'

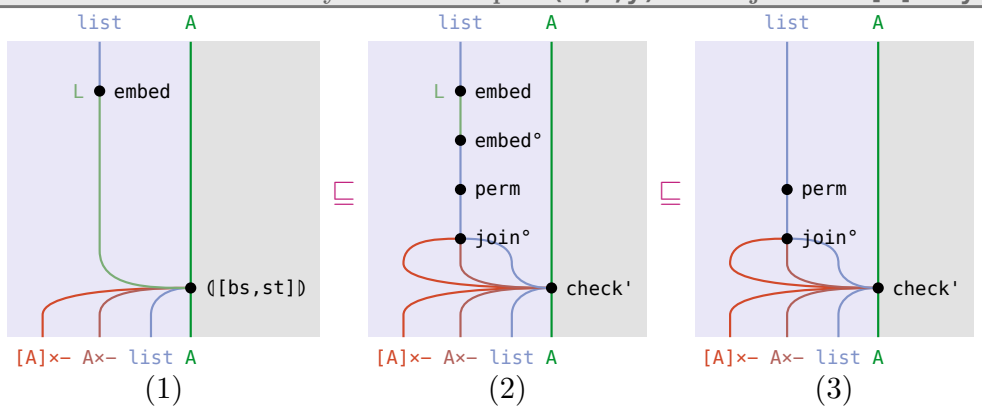


4	= (flatten × (𝟙 × flatten)) check' (perm × (𝟙 × perm)) join perm	(6.6.3d) join perm = (perm × (𝟙 × perm)) join perm
5	= (flatten × (𝟙 × flatten)) check' join perm	(6.6.3c) check (flatten × (𝟙 × flatten)) = (flatten × (𝟙 × flatten)) check'
6	= check (flatten × (𝟙 × flatten)) join perm	fork flatten = (flatten × (𝟙 × flatten)) join
7	= check fork flatten perm	

`bs ⊑ τ perm join° check'; (𝟙 × (perm join° check')) st ⊑ cons perm join° check'` (6.6.3g)

`⇒ embed (|[bs,st]|) ⊑ perm join° check'`

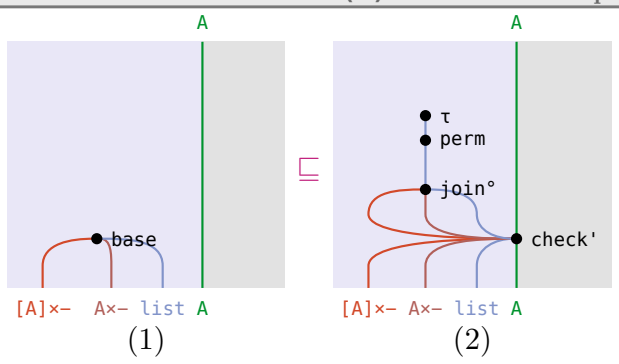
turning a list into a non-empty list and folding it with any `[bs, st]` meeting the two conditions below gives only **check**ed triples `(x,a,y)` whose join `x ⧺ [a] ⧺ y` is a permutation of the list



1	`embed (|[bs,st]|)`	`bs ⊑ τ perm join° check'; (𝟙 × (perm join° check')) st ⊑ cons perm join° check' ⇒ embed (|[bs,st]|) ⊑ embed embed° perm join° check'`
2	`⊑ embed embed° perm join° check'`	`embed embed° perm join° check' ⊑ perm join° check'`
3	`⊑ perm join° check'`	

`base ⊑ τ perm join° check'` (6.6.3h)

`base(a)` is a **check**ed triple whose join is a permutation of the one-element list `[a]`

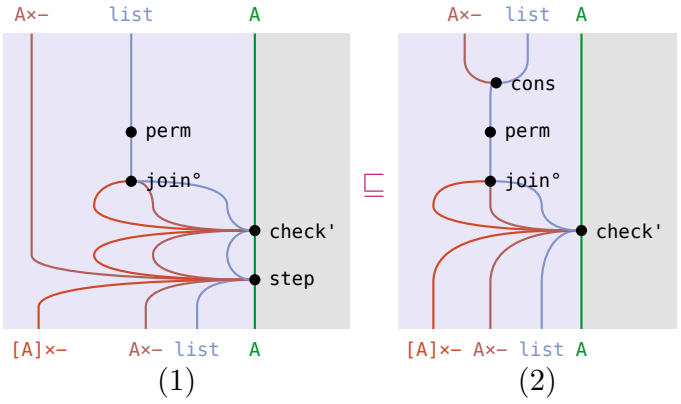


1	`base`	`[] ⧺ [a] ⧺ [] = [a]`, and `check'` holds on empty lists
2	`⊑ τ perm join° check'`	

`connected(R)` (6.6.3i)

`⇒ (𝟙 × (perm join° check')) step ⊑ cons perm join° check'`

if **`(x,b,y)`** is a **check**ed triple for a permutation of **`l`**, then **`step(a,(x,b,y))`** is a **check**ed triple for a permutation of **`[a] ⧺ l`**

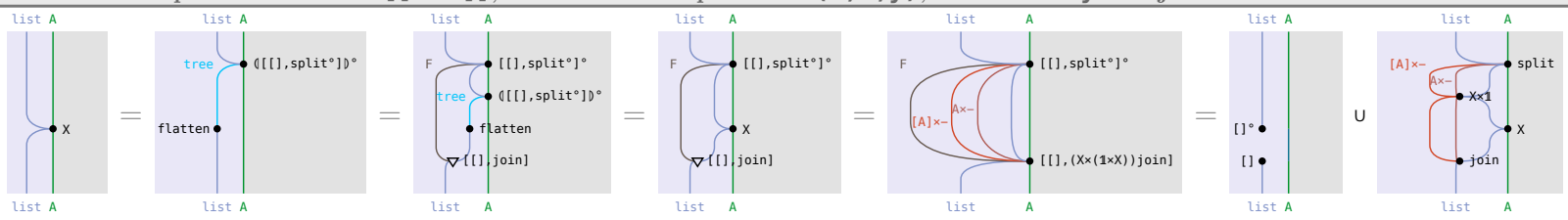


1	`(𝟙 × (perm join° check')) step`	`aRb` puts `a` in front of `x`; otherwise `bRa` and `a` goes in front of `y`
2	`⊑ cons perm join° check'`	

X = ◖[[[],split°]◗° flatten (6.6.3j)

⇒ X = ([]° []) ∪ (split (X × (𝟙 × X)) join)

quicksort returns **[]** on **[]**, and otherwise splits into **(x,a,y)**, sorts **x** and **y** and joins them around **a**



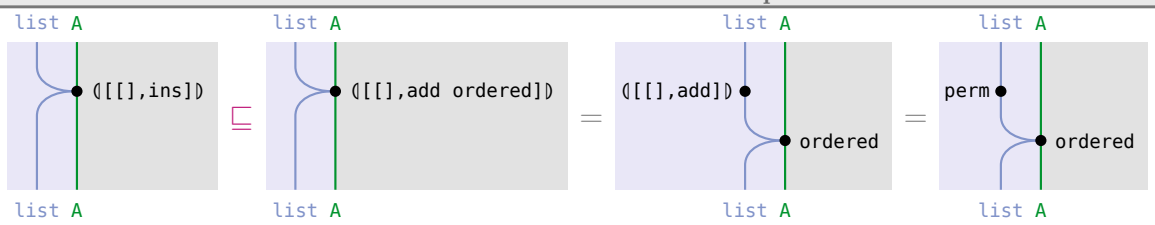
(1) (2) (3) (4) (5) (6)

1	X	X = ◖[[[],split°]◗° flatten
2	= ◖[[[],split°]◗° flatten	T° F(H) h = H
3	= [[],split°]° F(◖[[[],split°]◗° flatten) [[],join]	X = ◖[[[],split°]◗° flatten
4	= [[],split°]° F(X) [[],join]	F(X) [T,join] = [T,(X × (𝟙 × X)) join]
5	= [[],split°]° [[],(X × (𝟙 × X)) join]	[U,V]° [R,S] = (U° R) ∪ (V° S)
6	= ([]° []) ∪ (split (X × (𝟙 × X)) join)	

(𝟙 × ordered) ins ⊑ add ordered (6.6.3k)

⇒ ◖[[[],ins]◗ ⊑ perm ordered

for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with **[nil, ins]** gives only sorted permutations



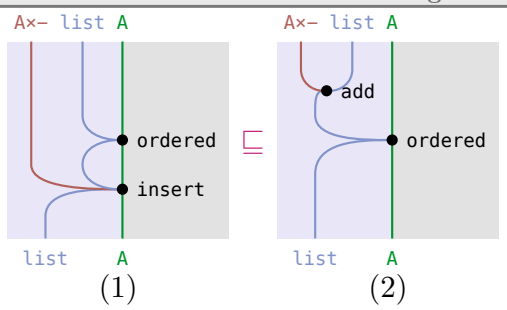
(1) (2) (3) (4)

1	◖[[[],ins]◗	(𝟙 × ordered) ins ⊑ add ordered ⇒ ◖[[[],ins]◗ ⊑ ◖[[[],add ordered]◗
2	⊑ ◖[[[],add ordered]◗	◖[[[],add]◗ ordered = ◖[[[],add ordered]◗
3	= ◖[[[],add]◗ ordered	perm = ◖[[[],add]◗
4	= perm ordered	

connected(R); (∀a. ∀b. ∀c. R(a,b) ⇒ R(b,c) ⇒ R(a,c)) (6.6.3l)

⇒ (𝟙 × ordered) insert ⊑ add ordered

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(1) (2)

1	(𝟙 × ordered) insert	insert splices a in, and R transitive and connected
2	⊑ add ordered	

§6.7. Closure

$$R^* = (\mu X : \mathbb{1} \cup (X\ R)) \tag{6.7a}$$

closing R by composing it on the left and closing it by composing on the right give the same relation R^*

$$\mathbb{1} \sqsubseteq X;\ X^2 \sqsubseteq X \Rightarrow R \sqsubseteq X \Leftrightarrow R^* \sqsubseteq X \tag{6.7b}$$

X contains R exactly when it contains R^*

$$\mathbb{1} \sqsubseteq R^* \tag{6.7c}$$

R^* is reflexive

1	$\mathbb{1}$	$R \sqsubseteq R \cup S$
2	$\sqsubseteq\ \mathbb{1} \cup (R\ R^*)$	$\mathbb{1} \cup (R\ R^*) = R^*$
3	$=\ R^*$	

$$R \sqsubseteq R^* \tag{6.7d}$$

R^* contains R

1	R	
2	$=\ R$	(6.7c) $\mathbb{1} \sqsubseteq R^*$
3	$\sqsubseteq\ R\ R^*$	$R\ R^* \sqsubseteq R^*$
4	$\sqsubseteq\ R^*$	

$$(\mathbb{1} \cup (R\ (R^* \,/\, R^*)))\ R^* \sqsubseteq R^* \tag{6.7e}$$

R^*/R^* is a prefixed point of $X \mapsto \mathbb{1} \cup RX$, stated through division

$$R^{*2} \sqsubseteq R^*$$

R^* is transitive, because R^*/R^* is a prefixed point of $X \mapsto \mathbb{1} \cup RX$

1	$(\mathbb{1} \cup (R\ (R^* \,/\, R^*)))\ R^*$	$(S \cup T)\ R = (S\ R) \cup (T\ R)$
2	$=\ (R^*) \cup (R\ (R^* \,/\, R^*)\ R^*)$	
3	$=\ R^* \cup (R\ (R^* \,/\, R^*)\ R^*)$	$(X \,/\, X)\ X \sqsubseteq X$
4	$\sqsubseteq\ R^* \cup (R\ R^*)$	$R\ R^* \sqsubseteq R^*$
5	$\sqsubseteq\ R^*$	

$$\mathbb{1} \sqsubseteq X;\ X^2 \sqsubseteq X;\ R \sqsubseteq X \Rightarrow \mathbb{1} \cup (R\ X) \sqsubseteq X$$

a reflexive transitive X containing R is a prefixed point of $Y \mapsto \mathbb{1} \cup RY$

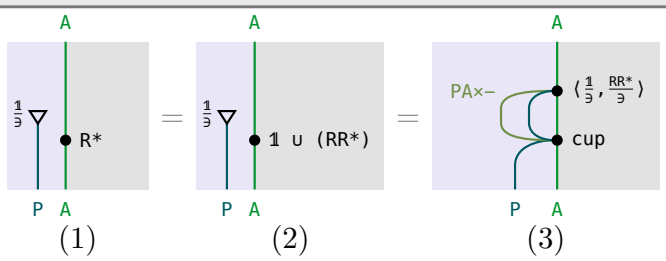
$$\mathbb{1} \sqsubseteq X;\ X^2 \sqsubseteq X;\ R \sqsubseteq X \Rightarrow R^* \sqsubseteq X \tag{6.7f}$$

a reflexive transitive X containing R is a prefixed point of $X \mapsto \mathbb{1} \cup RX$, so it contains R^*

1	$\mathbb{1} \cup (R\ X)$	$\mathbb{1} \sqsubseteq X$
2	$\sqsubseteq\ X \cup (R\ X)$	$R \sqsubseteq X$
3	$\sqsubseteq\ X \cup (X^2)$	
4	$\sqsubseteq\ X$	

$$\frac{R^*}{\ni} = \langle \frac{\mathbb{1}}{\ni}, \frac{R\ R^*}{\ni} \rangle\ \mathrm{cup} \tag{6.7g}$$

the set of R^*-successors of a is a itself joined with the R^*-successors of its R-successors



1	$\frac{R^*}{\ni}$	$\mathbb{1} \cup (R\ R^*) = R^*$
2	$=\ \frac{\mathbb{1} \cup (R\ R^*)}{\ni}$	$\frac{R \cup S}{\ni} = \langle \frac{R}{\ni}, \frac{S}{\ni} \rangle\ \mathrm{cup}$
3	$=\ \langle \frac{\mathbb{1}}{\ni}, \frac{R\ R^*}{\ni} \rangle\ \mathrm{cup}$	

§6.7.1. Computing closure

R − ∅ = R (6.7.1a)

taking nothing away leaves R

R ∪ (S − R) = R ∪ S

beside R, only the part of S outside R adds anything

R − (S ∪ T) = R − S − T

taking away S and then T takes away S∪T

(R ∪ S) − T = (R − T) ∪ (S − T)

taking T away from a union takes it away from each part

monotonic(φ); monotonic(ψ) (6.7.1b)

⇒ (μX : φ(ψ(X))) = φ(μY : ψ(φ(Y)))

the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ

S R* = (μX : S ∪ (X R)) (6.7.1c)

S followed by any number of R steps is the least X containing S and closed under a further R step

R* S = (μX : S ∪ (R X))

any number of R steps followed by S is the least X containing S and closed under an R step in front

θ(∅,S) = S R* (6.7.1d)

started with nothing found and S to explore, θ computes SR*

1	θ(∅,S)	θ(P,S) ≜ P ∪ (μX : S ∪ ((X R) − P))
2	≜ ∅ ∪ (μX : S ∪ ((X R) − ∅))	(6.7.1a) R − ∅ = R
3	= ∅ ∪ (μX : S ∪ (X R))	∅ ∪ R = R
4	= (μX : S ∪ (X R))	(6.7.1c) S R* = (μX : S ∪ (X R))
5	= S R*	

θ(P,∅) = P (6.7.1e)

with nothing left to explore, θ returns what it has found

1	θ(P,∅)	θ(P,Q) ≜ P ∪ (μX : Q ∪ ((X R) − P))
2	≜ P ∪ (μX : ∅ ∪ ((X R) − P))	(μX : ∅ ∪ ((X R) − P)) = ∅
3	= P ∪ ∅	P ∪ ∅ = P
4	= P	